Creation of a Node – Singly Linked List

```
#include <stdio.h>
#include <stdlib.h>

/* Node structure */
struct Node
{
    int data;
    struct Node *next;
};

/* Function to create a node */
struct Node* createNode(struct Node *head)
{
    struct Node *newnode, *temp;
    int value;

    newnode = (struct Node*)malloc(sizeof(struct Node));

    printf("Enter data: ");
    scanf("%d", &value);

    newnode->data = value;
    newnode->next = NULL;

    if(head == NULL)
    {
        head = newnode;
    }
    else
    {
        temp = head;

        while(temp->next != NULL)
        {
            temp = temp->next;
        }

        temp->next = newnode;
    }

    return head;
}
```

```c
/* Function to display the linked list */
void display(struct Node *head)
{
    struct Node *temp;

    if(head == NULL)
    {
        printf("List is empty\n");
        return;
    }

    temp = head;

    printf("Linked List: ");

    while(temp != NULL)
    {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }

    printf("NULL\n");
}

/* Main function */
int main()
{
    struct Node *head = NULL;
    int n, i;

    printf("Enter number of nodes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        head = createNode(head);
    }

    display(head);

    return 0;
}
```

Example Output

```
Enter number of nodes: 3
Enter data: 10
Enter data: 20
Enter data: 30

Linked List: 10 -> 20 -> 30 -> NULL
```

Functions Used

Function	Purpose
createNode()	Creates and adds a node to the list
display()	Prints the linked list
main()	Controls program execution